City of Santa Clara

Water Quality Table

	UNIT	MCL	State PHG/ Fed MCLG	Analysis for CSC WELL WATER Range	Analysis for CSC WELL WATER Average	Analysis for VALLEY WATER Range	Analysis for VALLEY WATER Average	Analysis for SFPUC Range	Analysis for SFPUC Average [highest running average]	Common Sources
Primary Standards For Source Water Sampling										
MICROBIOLOGICAL										
Giardia lamblia	cyst/L	TT	0	NA	NA	ND - 0.1	ND	0 - 0.05	0.01	naturally present in environment
RADIOACTIVITY										
Gross Alpha	pCi/L	15	(0)	NA	NA	3.3	3.3	ND	ND	erosion of natural deposits
Uranium	pCi/L	20	0.43	NA	NA	1.3	1.3	ND	ND	erosion of natural deposits
INORGANIC CHEMICAL										
Barium	PPM	1	2	ND - 0.21	0.12	ND	ND	ND	ND	erosion of natural deposits/oil drilling
Fluoride	PPM	2	1	0.12 - 0.18	0.15	ND - 0.1	ND	0.5 - 0.8 (1)	0.7	water additive/erosion of natural deposits
Hexavalent Chromium	PPB	10	0.02	0.78 - 2.6	1.6	ND	ND	ND - 0.1	ND	natural deposit leaching
Nitrate (as Nitrogen)	PPM	10	10	ND - 5.6	3.1	ND - 0.6	ND	ND - 0.4	ND	erosion of natural deposits/runoff/leaching
Nitrate & Nitrite (as Nitrogen)	PPM	10	10	2.3 - 5.6	3.9	ND - 0.6	ND	NA	NA	naturally present in environment
Perchlorate	PPB	6	1	ND - 1.4	ND	ND	ND	ND	ND	environmental contamination
Secondary Standards: Consumer Acceptance Contaminent Levels										
Aluminum (2)	PPB	200	600	ND - 30	13	ND	ND	ND - 59	ND	erosion of natural deposits; some surface water treatment residue
Chloride	PPM	500	NA	30 - 68	54	48 - 67	56	< 3 - 11	5.4	runoff/natural deposit leaching/seawater
Color	UNITS	15	NA	ND - 2	ND	ND - 2.0	ND	< 5	< 5	naturally occuring organic material
Iron	PPB	300	NA	ND - 200	45	< 20	< 20	< 6 - 36	18	natural deposit leaching/industrial wastes
Odor	UNITS	3	NA	ND - 2	ND	ND - 1.0	ND	ND	ND	naturally occuring organic material
Manganese	PPB	50	NA	ND - 2.9	2.4	< 1.0	< 1.0	< 2 - 2.7	< 2	natural deposit leaching
Foaming Agents (MBAS)	PPB	500	NA	ND - 0.022	0.014	< 0.10	< 0.10	ND	ND	municipal and industrial waste discharges
Specific Conductance	uS/cm	1600	NA	568 - 723	631	434 - 466	449	32- 346	189	substances that form ions when in water/ seawater intrusion
Sulfate	PPM	500	NA	36 - 54	43	50 -63	57	1 - 45	23	runoff/natural deposit leaching/industrial wastes
Total Dissolved Solids	PPM	1000	NA	232 - 666	450	254 - 270	261	24 - 197	111	runoff/natural deposit leaching
Turbidity	NTU	5	NA	0.36 - 2	1.1	0.01 - 0.47	0.04	< 0.1 - 0.3	0.1	soil runoff
Consumer Information										
pH	UNITS	NS	NS	7.6 - 7.9	7.7	7.4 - 7.9	7.6	NA	NA	
Alkalinity (as $CaCO_3$)	PPM	NS	NS	187 - 255	227	65 - 71	68	8 - 131	61	
Bicarbonate Alkalinity (as HCO_3)	PPM	NS	NS	NA	NA	< 10 - 87	63	ND	ND	
Boron	PPB	NS	NS	NA	NA	120- 152	130	21 - 71	46	
Calcium (as Ca)	PPM	NS	NS	68 - 91	83	19.8 - 20.4	20.2	3.1 - 29	16	
Chlorate (3)	PPB	NS	NS	NA	NA	NA	NA	< 20 - 281	88	
Hardness	PPM	NS	NS	250 -330	293	96 - 98	97	8.1 - 112	60	
Magnesium	PPM	NS	NS	18 -24	21	11.1 - 11.8	11.4	0.2 - 10	5.1	
Phosphate	PPM	NS	NS	NA	NA	1.0 - 1.2	1.1	ND	ND	
Potassium	PPM	NS	NS	1.2 - 1.4	1.3	2.5 - 2.8	2.6	ND	ND	
Silica	PPM	NS	NS	NA	NA	10 - 14	12	5.3 - 7.8	6.6	
Sodium	PPM	NS	NS	23 - 39	29	50 - 56	52	3.1 - 29	16	
Temperature	°C	NS	NS	11.8 - 26.6	18.7	14.1 - 22.0	18.7	NA	NA	
Total Organic Carbon	PPM	NS	NS	NA	NA	1.3 - 2.0	1.7	1.4 - 3.1 (4)	2.1	
Vanadium	PPB	NS	NS	NA	NA	1 - 2	2	NA	NA	

All Santa Clara water sources meet or exceed Federal and State standards.

Primary Standards As Measured In City Of Santa Clara Distribution System

	UNITS	MCL	State MCL	Range	Average		Common Sources
MICROBIOLOGICAL							
Total Coliform	% pos (+)	5.0%	5.0%	0 - 3%	< 5.0%		naturally present in environment
Fecal Coliform and *E.coli* (5)	# of pos (+)	0	0	0	0		human and animal fecal waste
DISINFECTION BYPRODUCTS, RESIDUALS, PRECURSORS							
Trihalomethanes	PPB	80	NA	ND - 44.0	[43.3]		byproduct of drinking water disinfection
Haloacetic Acids	PPB	60	NA	ND - 48.1	[36.4]		byproduct of drinking water disinfection
Chlorine residual	PPM	4	4	0.0 - 4.6	0.59		drinking water disinfectant
INORGANIC CHEMICAL as measured at 52 Residential Taps in 2022:							
Copper	PPM	AL = 1.3	0.3	90th percentile = 0.34 ppm		Number Exceeded = 0	corrosion of plumbing systems
Lead	PPB	AL = 15	0.2	90th percentile = ND		Number Exceeded = 0	corrosion of plumbing systems
SCHOOLS REQUESTING LEAD TESTING IN 2018: 33 Schools (172 samples taken)							
Lead	PPB	AL = 15	0.2	90th percentile = ND		Number Exceeded = 1 (6)	corrosion of plumbing systems

Unregulated Contaminants As Measured In City Of Santa Clara Distribution System

	UNITS	Notification Level	Range	Average
Lithium	PPB	NA	ND - 10	ND

Definitions

DISINFECTION BYPRODUCTS = chemical byproducts of disinfection

DISTRIBUTION SYSTEM = drinking water delivery system

HARDNESS = the sum of polyvalent cations present in the water, generally magnesium and calcium. The cations are usually naturally occurring.

MAXIMUM CONTAMINANT LEVEL (MCL) = The highest level of a contaminant allowed in drinking water. Primary MCLs are set as close to the PHGs (or MCLGs) as is economically and technologically feasible. Secondary MCLs are set to protect the odor, taste, and appearance of drinking water.

MAXIMUM CONTAMINANT LEVEL GOAL (MCLG) = The level of a contaminant in drinking water below which there is no known or expected risk to health. MCLGs are set by the U.S. EPA.

MAXIMUM RESIDUAL DISINFECTANT LEVEL (MRDL) = The highest level of a disinfectant allowed in drinking water. There is convincing evidence that addition of a disinfectant is necessary for control of microbial contaminants.

MAXIMUM RESIDUAL DISINFECTANT LEVEL GOAL (MRDLG) = The level of a drinking water disinfectant below which there is no known or expected risk to health. MRDLGs do not reflect the benefits of the use of disinfectants to control microbial contaminants.

NA = not applicable or available

ND = not detected

NS = no standard

NTU = Nephelometric Turbidity Unit. Turbidity is a measure of the cloudiness of the water. We monitor it because it is a good indicator of water quality.

pCi/L = picocuries per liter (a measure of radioactivity)

PPB = Parts Per Billion (µg/L)

PPM = Parts Per Million (mg/L)

PRIMARY DRINKING WATER STANDARDS (PDWS) = MCLs and MRDLs for contaminants that affect health along with their monitoring and reporting requirements, and water treatment requirements.

PUBLIC HEALTH GOAL (PHG) = The level of a contaminant in drinking water below which there is no known or expected risk to health. PHGs are set by the California EPA.

REGULATORY ACTION LEVEL (AL) = The concentration of a contaminant which, if exceeded, triggers treatment or other requirements that a water system must follow.

RESIDENTIAL TAPS = household faucets used for lead and copper sampling

SECONDARY STANDARDS = secondary MCLs are set to protect the aesthetics of drinking water

SODIUM = refers to the salt present in the water and is generally naturally occurring.

TREATMENT TECHNIQUE (TT) = A required process intended to reduce the level of a contaminant in drinking water.

TURBIDITY= Very small/micrscopic particles in water.

UNREGULATED CONTAMINANTS = Unregulated contaminant monitoring helps EPA and SWRCB to determine where certain contaminants occur and whether the contaminants need to be regulated. Every four years, the EPA updates the list of unregulated contaminants to monitor.

Notes

(1) Natural fluoride in the Hetch Hetchy water was ND. The fluoride level in treated water ranged from 0.5 to 0.8 ppm with an average of 0.7 ppm.

(2) Aluminum also has a primary MCL of 1,000 ppb.

(3) The detected chlorate in the treated water is a degradation product of sodium hypochlorite used by the San Francisco Regional Water System for water disinfection.

(4) The range and average values of the total organic carbon were from operational monitoring results at Alameda East and SVWTP effluent.

(5) The MCL was changed to *E. coli* starting on July 1, 2021 when the State Revised Total Coliform Rule became effective.

(6) 2018 sampling result for John Sutter Elementary was 26 ppb. Repeat sampling following plumbing repairs was non-detect for lead.